and apprehend the now confirmed terrorist possibly preventing a number of future crimes and saving lives.

Amazingly, the helm of the operation, the person responsible for identification of the criminal is not a human at all, it is in fact, a computer program tasked with monitoring the secure CCTV channels across the city and looking for known criminals.

Sounds like the stuff of fiction? Welcome to Moscow, the year is 2017.

The last decade saw a new technique of computer programming emerge that would take the world by storm. Instead of programming a computer to specifically process a set of data in a certain way, what if we could show it a large set of input and outputs so that it is able to figure out (with some degree of approximation) a programming logic that would generate those results given the inputs? That is the essence of machine learning.

Unlike classical programming approaches, machine learning does not seek to explicitly determine the logic flow of the application and instead uses iterative mathematical optimisations to determine a near approximate function, known as a model that converts the inputs supplied to the mapped outputs. This model is then used to evaluate real life inputs with the assumption that it would generate an output that is close to the actual output.

A detailed study on machine learning is out of the scope of the tutorial but we will try to clarify most terminologies and processes as we go through our example. However, a basic understanding of machine learning and mathematics would definitely help the reader to better utilize the offerings on AWS.

Amazon Machine Learning (ML)

Amazon Machine Learning (Amazon ML), lets you build and train predictive models and host your applications in a scalable environment. It also allows you to process input datasets for better optimization of training.

Getting started with Amazon ML:

- To start creating your own model on Amazon ML, go to the Services tab and open 'Amazon Machine Learning' service from it. Click on the 'Get Started Now' button and you will be redirected to a standard setup wizard.
- First, you have to add training data for your model, to do this add your data to a S3 bucket and provide the file URL in the S3 data section on the input page. Alternatively you can also specify Redshift as a datasource for your model.